

Precision and Recall in Machine Learning

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Precision and recall are two evaluation metric used to **check the performance of Machine Learning Model**. **Precision** is the ratio of a model's classification of all positive classifications as positive. **Recall** tells us how many of the actual positive items the model was able to find. Precision and recall helps in classification problems. In this article we will explain Precision and Recall with formula and example.

1. Precision

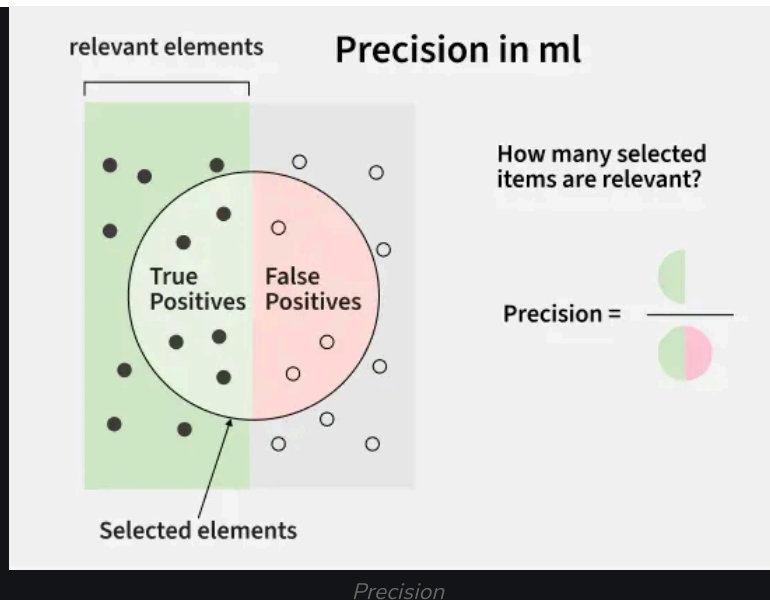
Precision is the ratio between the True Positives and all the Positives. It shows how many of the "yes" predictions made by the model were actually correct. It helps us reduce wrong "yes" guesses which are called **false positives (FP)**. Precision is calculated as:

$$\text{Precision} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Positives (FP)}}$$

Precision

Imagine you build a model to find **birds in photos**. It marks some photos as "bird."

- If those marked photos really have birds that's good (true positives).
- But if some don't have birds the model made a mistake (false positives).



Uses of Precision

- **Precision** helps us understand how accurate a model's "yes" predictions are. It is especially useful when the data has more of one kind of result than the other.
- For example if most emails are not spam and only a few are then precision helps us see how well the model is finding the spam without making too many mistakes. In such uneven data precision helps measure how correctly the model is picking out the less common group like spam or fraud.

Advantages of High Precision

A model with high precision is very good at avoiding mistakes when it says "yes." This is important in situations where false alarms are a big problem. For example:

- In spam email detection it's better if real emails don't get wrongly marked as spam.
- We care more about getting the important emails right than stopping every single spam message.

So in these cases a model that gives fewer wrong "yes" answers is more useful.

Limitations of Precision

- If we only care about precision then model may miss some real cases. It becomes too careful and may say “no” even when something is actually “yes.”
- If the model is too focused on being precise it might let lots of spam emails into your inbox because it's afraid of wrongly marking a real email as spam.

2. Recall

Recall tells us how well a model finds all the correct “yes” cases in the data. It checks **how many real positive cases** the model was able to correctly identify. The formula to calculate recall is:

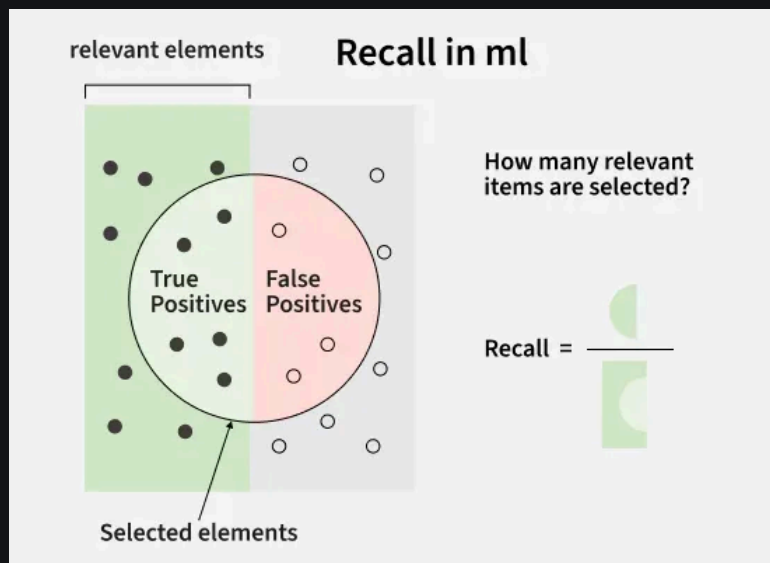
$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$$

Recall

- **True Positives (TP)**: The model correctly said “yes.”
- **False Negatives (FN)**: The model missed a real “yes” and said “no” instead.

Imagine a computer model that looks for birds in pictures.

- Recall tells us how many real birds the model found correctly.
- A perfect model would find all birds with no misses that means no false negatives.



Uses of Recall

You use recall when it's very important to find all possible positive cases even if some of them turn out to be wrong. For example:

- In medical tests you want to catch every possible patient who may be sick even if that means a few healthy people are wrongly flagged.
- In fraud detection it's better to check a few extra normal transactions than to miss a real fraud.

Advantages of High Recall

A model with high recall is very good at not missing anything important. It finds almost all the actual "yes" cases in the data. This is helpful when:

- Missing a real case is dangerous or costly.
- For example in cybersecurity missing an attack is worse than accidentally flagging something safe.

Limitations of Recall

Focusing only on recall means the model is optimized to identify as many actual positives as possible even at the cost of incorrectly labeling negatives as positives. This often leads to a high number of false positives.

Combining Precision and Recall (F1 Score)

F1 score is a metric that combines **precision** and **recall** into one value allow to optimize both at the same time. Precision measures how many of the predicted positives are actually correct while recall measures how many actual positives the model correctly identified. F1 score is calculated using the formula:

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

This formula provides a balance between the two metrics. The F1 score is particularly useful when you have imbalanced data meaning one class is much larger than the other. A perfect F1 score of 1.0 indicates both precision and recall are perfect whereas a score of 0.0 suggests poor performance. By maximizing F1 score you can ensure that the model performs well in both avoiding false positives and missing true positives.

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